

Proof that $\sqrt{7}$ is Irrational

Problem

Prove that $\sqrt{7}$ is irrational.

Proof

Lemma 1. *Let $n \in \mathbb{Z}$. If $7 \mid n^2$, then $7 \mid n$.*

Proof. By the division algorithm, applied with positive divisor 7, there exist $q \in \mathbb{Z}$ and $r \in \{0, 1, 2, 3, 4, 5, 6\}$ such that

$$n = 7q + r.$$

We first justify explicitly that n^2 and r^2 have the same remainder modulo 7. Indeed,

$$n^2 - r^2 = (7q + r)^2 - r^2 = 49q^2 + 14qr = 7(7q^2 + 2qr).$$

Since $7q^2 + 2qr \in \mathbb{Z}$, this shows that

$$7 \mid (n^2 - r^2).$$

Now assume $7 \mid n^2$. Since both n^2 and $n^2 - r^2$ are divisible by 7, their difference is also divisible by 7. Hence

$$r^2 = n^2 - (n^2 - r^2)$$

is divisible by 7.

We now check the possible values of r :

r	0	1	2	3	4	5	6
r^2	0	1	4	9	16	25	36
$r^2 \bmod 7$	0	1	4	2	2	4	1

Thus, among $r \in \{0, 1, 2, 3, 4, 5, 6\}$, the condition $7 \mid r^2$ holds only when $r = 0$. Therefore $r = 0$, and so

$$n = 7q.$$

Hence $7 \mid n$. □

Theorem 2. $\sqrt{7}$ is irrational.

Proof. Suppose, for contradiction, that $\sqrt{7}$ is rational. Then there exist integers $m, n \in \mathbb{Z}$ with $n \neq 0$ such that

$$\sqrt{7} = \frac{m}{n}.$$

Since $n \neq 0$, either $n > 0$ or $n < 0$. If $n < 0$, replace m and n by $-m$ and $-n$. This does not change the fraction because

$$\frac{-m}{-n} = \frac{m}{n},$$

and the new denominator $-n$ is positive. Therefore we may assume

$$n > 0.$$

Let

$$g = \gcd(m, n).$$

Since $n \neq 0$, the integers m and n are not both zero, so g is a positive integer. In particular,

$$g \neq 0.$$

Because $g \mid m$ and $g \mid n$, there exist integers $a, b \in \mathbb{Z}$ such that

$$m = ga, \quad n = gb.$$

Since $n > 0$ and $g > 0$, we have $b > 0$, so in particular $b \neq 0$. Therefore

$$\sqrt{7} = \frac{m}{n} = \frac{ga}{gb} = \frac{a}{b},$$

where the cancellation is valid because $g \neq 0$.

We claim that $\gcd(a, b) = 1$. Let d be any positive integer dividing both a and b . Then $a = du$ and $b = dv$ for some integers u, v . Hence

$$m = ga = gdu, \quad n = gb = gdv,$$

so gd is a positive common divisor of m and n . Since $g = \gcd(m, n)$ is the greatest positive common divisor of m and n , we must have

$$gd \leq g.$$

Because $g > 0$, dividing by g gives $d \leq 1$. Since d is a positive integer, $d = 1$. Thus the only positive common divisor of a and b is 1, so

$$\gcd(a, b) = 1.$$

Now square both sides of $\sqrt{7} = a/b$. Since equality is preserved under squaring,

$$(\sqrt{7})^2 = \left(\frac{a}{b}\right)^2.$$

By the definition of $\sqrt{7}$, we have $(\sqrt{7})^2 = 7$. Also, since $b \neq 0$,

$$\left(\frac{a}{b}\right)^2 = \frac{a^2}{b^2}.$$

Therefore $7 = a^2/b^2$. Multiplying both sides by b^2 gives

$$a^2 = 7b^2.$$

Hence $7 \mid a^2$. Since $a \in \mathbb{Z}$, Lemma 1 applies, and we get $7 \mid a$. Thus $a = 7k$ for some $k \in \mathbb{Z}$. Substituting into $a^2 = 7b^2$:

$$(7k)^2 = 7b^2 \implies 49k^2 = 7b^2 \implies 7k^2 = b^2.$$

Thus $7 \mid b^2$. Since $b \in \mathbb{Z}$, Lemma 1 applies again, and therefore $7 \mid b$.

We have shown that $7 \mid a$ and $7 \mid b$. Hence 7 is a positive common divisor of a and b , with $7 > 1$, contradicting $\gcd(a, b) = 1$.

Therefore our assumption that $\sqrt{7}$ is rational is false. Hence $\sqrt{7}$ is irrational.

□